$M = \{ A = [a_{ij}]_{m \times n} \}$

$\langle A, B \rangle = tr(B^T A)$

I1 $\langle \alpha A_1 + \beta A_2, B \rangle = tr(B^T(\alpha A_1 + \beta A_2))$

$= tr((B^T \alpha A_1) + (B^T \beta A_2))$

$= tr(\alpha B^T A_1 + \beta B^T A_2)$

$= tr(\alpha B^T A_1) + tr(\beta B^T A_2)$

$= \alpha tr(B^T A_1) + \beta tr(B^T A_2)$

$= \alpha \langle A_1, B \rangle + \beta \langle A_2, B \rangle$

I2) $\langle A, B \rangle = tr(B^T A) = tr(B^T A)^T$

$= tr(A^T B)$

$= \langle B, A \rangle$

I3) Let $\langle A, A \rangle =$

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